

Raw Data Description

Trade, Migration and Productivity: A Quantitative Analysis of China
Replication Files (20150811_backup)

1. migration_data.dta

Full province-sector matrix of migration shares, bilateral distances, and instrumental variables
Shape: 3,540 rows x 14 columns

Variable	Description
importer	Province name (destination)
exporter	Province name (origin)
j	Sector code (1=agriculture, 2=non-agriculture)
i	Sector code (1=agriculture, 2=non-agriculture)
mij_mii2000	Migration share relative to staying in 2000
mij_mii2005	Migration share relative to staying in 2005
Vj	Real income in destination province-sector in 2000
Vi	Real income in origin province-sector in 2000
mij2000	Bilateral migration share in 2000
mij2005	Bilateral migration share in 2005
distance	Bilateral distance between provinces (km)
rYLj_bartik_sector	Bartik IV: weighted average of neighbor income by sector
rYLj_bartik_sector_uhs2000	Bartik IV: unweighted harmonic mean specification
rYLj_IVneighbour	Neighbor income as IV for migration elasticity

2. trade_data.dta

Bilateral trade flows between 8 regions for 2002 and 2007
Shape: 144 rows x 25 columns

Variable	Description
i	Origin region code
j	Destination region code
importer	Region name (destination)
exporter	Region name (origin)

Variable	Description
year	Year (2002 or 2007)
Eij_ag	Exports from i to j in agriculture
Eij_na	Exports from i to j in non-agriculture
Ei_ag	Total exports from i in agriculture
Ei_na	Total exports from i in non-agriculture
Eii_ag	Home consumption in agriculture
Eii_na	Home consumption in non-agriculture
Plij_ag	Trade share agriculture
Plij_na	Trade share non-agriculture
Plii_ag	Home share agriculture
Plii_na	Home share non-agriculture
ti_ag	Exporter-specific cost agriculture
ti_na	Exporter-specific cost non-agriculture
tj_ag	Importer-specific cost agriculture
tj_na	Importer-specific cost non-agriculture
tij_ag	Bilateral trade cost agriculture
tij_na	Bilateral trade cost non-agriculture

3. employment_realGDP_data.csv

Provincial real GDP per worker and employment in 2000 and 2005

Shape: 30 rows x 9 columns (30 provinces)

Variable	Description
province	Province name
rYLa2000	Real GDP per worker in agriculture, 2000
rYLn2000	Real GDP per worker in non-agriculture, 2000
La2000	Employment in agriculture, 2000 (millions)
Ln2000	Employment in non-agriculture, 2000 (millions)
rYLa2005	Real GDP per worker in agriculture, 2005
rYLn2005	Real GDP per worker in non-agriculture, 2005
drYLa_data	Observed change in agriculture real income
drYLn_data	Observed change in non-agriculture real income

4. mij2000.csv & mij2005.csv

Bilateral migration shares in vector form (to be reshaped in MATLAB)

Shape: 3,600 rows x 2 columns each

Variable	Description
mij_mii	Migration share relative to staying (mij/mii)
mij	Raw migration share (bilateral)
	Matrix structure: 60x60 (30 provinces x 2 sectors)
	Flattened: [prov1_ag, prov1_na, prov2_ag, ..., prov30_na]

5. trade_ag.csv & trade_na.csv

Bilateral trade matrix between 30 provinces in 2002

Shape: 30 rows x 30 columns each

Matrix Structure	30x30 bilateral trade flow matrix
Row	Exporter province
Column	Importer province
Value	Trade share (proportion)
Note	Provinces ordered by 'region' variable

6. tauhat.csv

Measured change in migration costs

Shape: 961 rows x 6 columns

Variable	Description
dni_ag	Change in migration cost (agriculture, symmetric)
dni_na	Change in migration cost (non-agriculture, symmetric)
dni_asym_ag	Change in asymmetric cost (agriculture)
dni_asym_na	Change in asymmetric cost (non-agriculture)
dni_dist_ag	Change in distance cost component (agriculture)
dni_dist_na	Change in distance cost component (non-agriculture)

7. dT.mat

Estimated productivity changes for model to match observed real income changes

Shape: MATLAB binary file

Content	Productivity changes for each province-sector
Dimensions	60x1 vector (30 provinces x 2 sectors)

Additional Information

Sector Codes	1 = Agriculture, 2 = Non-Agriculture
Years	Migration: 2000, 2005 Trade: 2002, 2007
Regions (8)	Central coast, Eastern coast, Far east, North, North coast, Northwest, Southcentral, Southwest
Provinces (30)	Anhui, Beijing, Chongqing, Fujian, Gansu, Guangdong, Guangxi, Guizhou, Hainan, Hebei, Heilongjiang, Henan, Hubei, Hunan, Jiangsu, Jiangxi, Jilin, Liaoning, Shaanxi, Shandong, Shanghai, Sichuan, Tianjin, Yunnan, Zhejiang

Note: mij2000.csv and mij2005.csv are vector forms that must be reshaped to 60x60 matrices in MATLAB.

Data Source: China National Bureau of Statistics

Citation: Tombe, T. and Zhu, X. (2015) 'Trade, Migration and Productivity: A Quantitative Analysis of China'